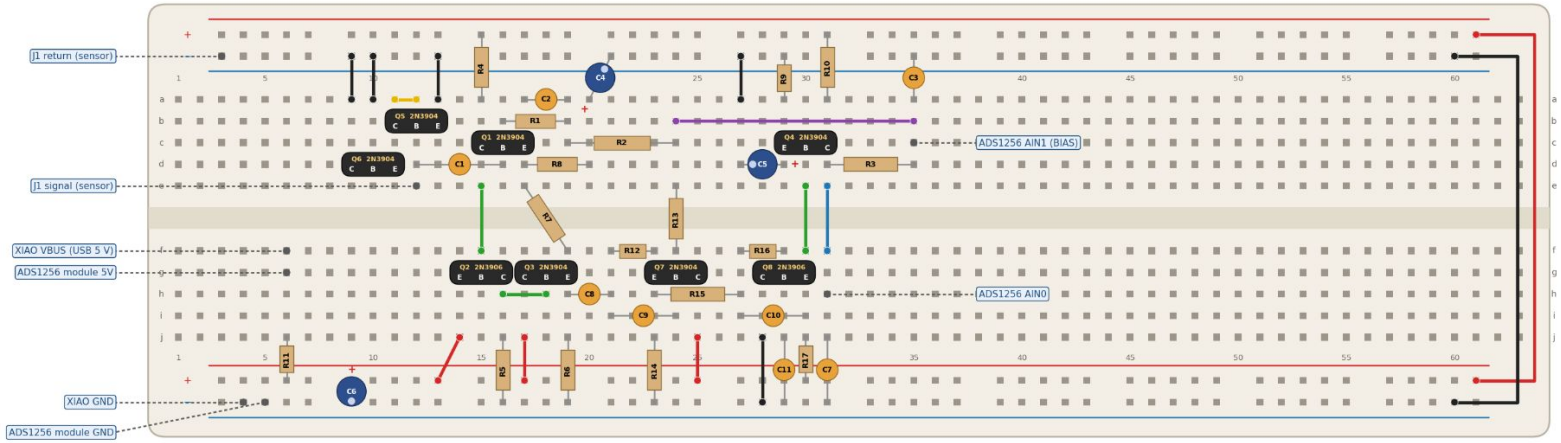
October 7th Progress Report

Ethan Ewer

Used Claude to build a breadboard layout for a circuit netlist

Receiver breadboard layout — standard 830-point board

Top view. Holes, parts, and jumpers match spice/receiver.cir node for node. Rails: top + and bottom + carry VA ≈ 4.8 V after R11; both − rails are GND.

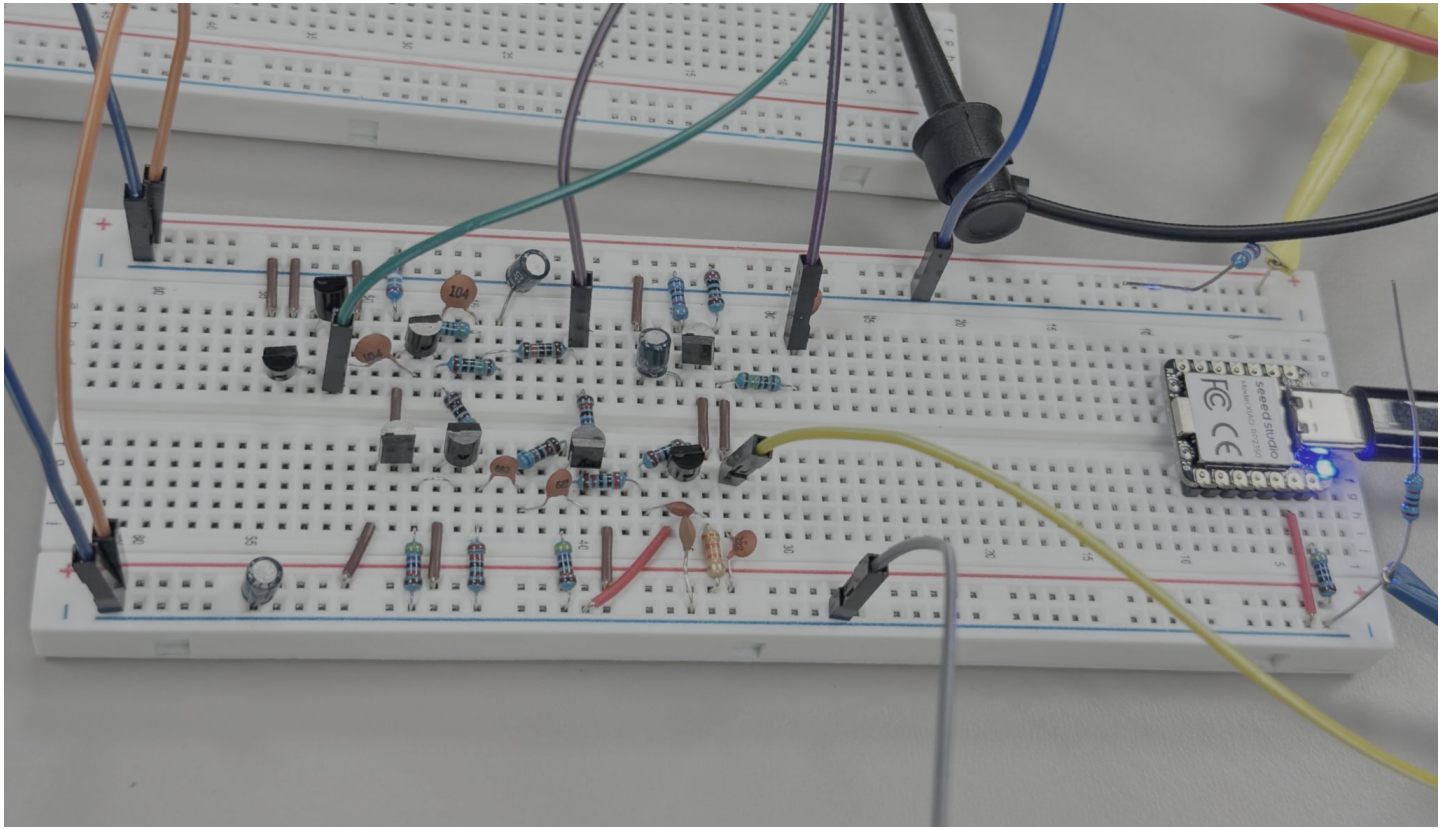


Part values

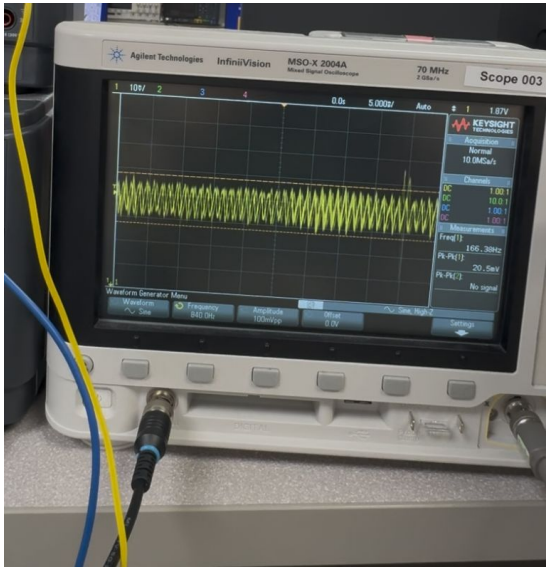
R1	470 kΩ	**R2**	100 kΩ	**R3**	1 MΩ	**R4**	24.9 kΩ	**R5**	47 kΩ	**R6**	2.2 kΩ	**R7**	5.1 kΩ	**R8**	150 Ω	**R9**	8.2 kΩ
R10	20 kΩ	**R11**	470 Ω	**R12**	6.8 kΩ	**R13**	47 kΩ	**R14**	47 kΩ	**R15**	20 kΩ	**R16**	20 kΩ	**R17**	33 kΩ	**C1**	100 nF
C2	100 nF	**C3**	100 nF	**C4**	1 µF	**C5**	1 µF	**C6**	470 µF	**C7**	2.2 nF	**C8**	6.8 nF	**C9**	6.8 nF	**C10**	6.8 nF
C11	1 nF	**Q1**	2N3904	**Q2**	2N3906	**Q3**	2N3904	**Q4**	2N3904	**Q5**	2N3904	**Q6**	2N3904	**Q7**	2N3904	**Q8**	2N3906

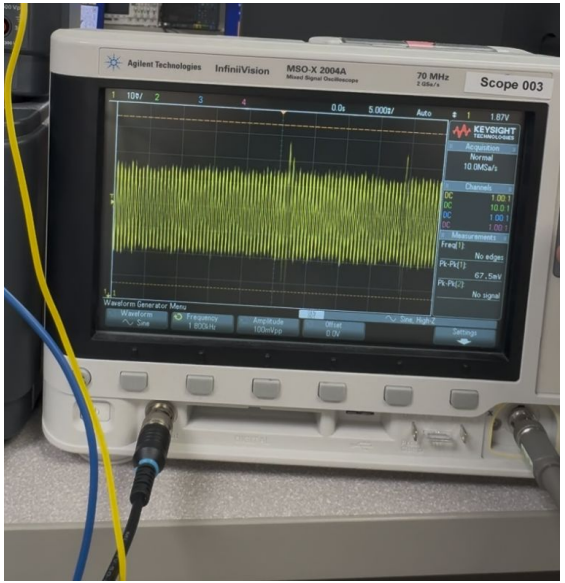
- Transistor letters give the lead each hole needs (E, B, C). Check every kit 2N3904/2N3906 pinout before inserting; do not assume E-B-C.
- Q5 and Q6 are clamp diodes: their B and C strips are joined (yellow jumper for Q5, GND jumpers for Q6). Electrolytics C4, C5, C6: the + mark is the positive lead.
- Rails are drawn continuous. If your board splits its rails at the middle, bridge each rail across the split. The red and black loops tie the top and bottom rail pairs.
- XIAO RP2350 and ADS1256 module stay off the board. Wire SPI directly between them: D8→SCLK, D10→DIN, D9←DOUT, D3→CS, D2←DRDY, D4→SYNC/PDWN.
- Keep J1 wiring short and away from the AIN0 wire; run the AIN0 and AIN1 wires together to the module. Solderless contacts add noise; measure the floor on the bench.

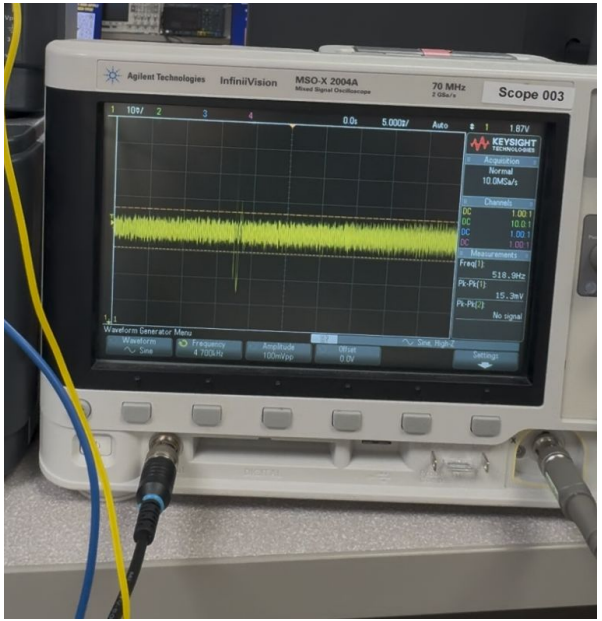
Assembled amplifier and band pass filter circuit



Measured amplifier gain as ~3600 V/V in passband, and lower outside







Next Steps

- Make circuit on a PCB
- Redesign the amplifier with op-amps instead of transistors (transistors were used because they were in the Thomson lab kits)
- Test amplifier with the rest of the system